

Akash Gogate

Princeton, NJ • 609-751-2597 • akash.gogate@gmail.com • agogate@wisc.edu • [linkedin.com/in/akashgogate](https://www.linkedin.com/in/akashgogate) • github.com/AkashGogate

EDUCATION

University of Wisconsin at Madison

Graduating: May 2027

Bachelor of Science in Computer Science • Dean's List Fall 2024, Spring 2025, Fall 2025 • 3.9 GPA

Relevant Coursework: Cryptography, Advanced Algorithms, Machine Organization and Programming, Intro to Computer Engineering, Data Structures, Artificial Intelligence, Deep Learning

South Brunswick High School

Relevant Coursework: Computer Science Capstone, Android Application Development, Virtual Reality, Game Development, Web Development

TECHNOLOGIES

Languages: Java, Python, TypeScript, Node.js, R, Rust, JavaScript, C#, C++, SQL, Linux/Bash

Systems & Data: Kafka, Kubernetes, Docker, MongoDB, SQLite, Redis, Supabase, REST APIs, RESTful Services, FastAPI, GraphQL, Low-Latency Systems, Financial Data Pipelines, Risk Engine, Regulatory Compliance, Event-Driven Architecture, Distributed Systems, Fault-Tolerant Systems, Time-Series Data

Frontend & Mobile: React.js, Android SDK, Spring Boot, HTML/CSS

Developer Tools: Git, GitHub, GitHub Actions, CI/CD, Linux, Version Control, pytest, Unit Testing, Agile, SDLC

Data & Analytics: NumPy, Pandas, PyTorch, Scikit-learn, SQL, NLP, RAG, Semantic Similarity Search, Statistical Modeling, Financial Modeling, Risk Analysis, Actuarial Data

EXPERIENCE

Leidos

May 2025 - Present

Software Engineer Intern

- Designed Kafka event-driven pipeline with at-least-once delivery semantics persisting to MongoDB, achieving **fault-tolerant, low-latency streaming** under strict operational and compliance constraints for mission-critical satellite data
- Optimized satellite scheduling via memoization and DP in Python; deployed containerized test suite on Kubernetes via **GitHub Actions CI/CD** with comprehensive unit coverage across all system components and integration tests
- Designed system architecture for decoupled, fault-tolerant data flow between scheduling, pipeline, and monitoring components; maintained data integrity and availability guarantees required by downstream satellite operations

The Kendziorz Lab at the University of Wisconsin-Madison

September 2025 - Present

Data / Research Engineer Intern

- Engineered automated data pipeline framework across **5+ scientific libraries**, reducing GPU processing costs by **300%** through systematic pipeline profiling, benchmarking, and data throughput optimization across the research stack
- Built TransferAgent module automating NLP data synthesis, cross-document scoring, and literature mapping workflows, cutting downstream processing time by **~60%** across the full pipeline and enabling faster research iteration

Inspirit AI Research Internship

Sep 2022 - Mar 2023

Machine Learning Research Intern

- Trained and tuned ML classifiers (Decision Tree, KNN) achieving **93% classification accuracy** via systematic hyperparameter tuning in scikit-learn with cross-validated statistical significance and robust evaluation methodology in a **model programming and tuning notebook**
- Published "**Early Skin Cancer Detection Improvement**" through Inspirit AI's peer-review process; separately trained a Random Forest on miRNA sequences achieving **95% accuracy**

TECHNICAL PROJECTS

Lotus Health (<https://github.com/AkashGogate/lotus-health>)

March 2026

- Built a financial risk engine computing 5-year disease cost risk paths across **45M financial records** on a **1,080-node ICD-10 comorbidity graph** via SQL queries over MEPS and CMS actuarial datasets; core output is a risk-adjusted, patient-level financial risk score
- Implemented RAG pipeline with semantic similarity search over graph-structured data as vector database for user financial risk scoring; NLP pipeline strictly separates language inference from deterministic financial logic, ensuring ACID-compliant, hallucination-free, and auditable cost outputs

Self-Improving LLM Agent (<https://github.com/AkashGogate/SelfImprovingLLMAgent>)

Jan 2026 - May 2026

- Failure-driven self-improvement loop for LLM agents across a **5,000+ task benchmark**, outperforming ReAct baseline by **15pp** via RAG-enhanced retrieval, Transformer reasoning, and inference-time optimization; provider-agnostic architecture with GitHub Actions CI/CD

LLM-Based SAT Practice Test Generator (<https://github.com/AkashGogate/SATPracticeTestGenerator>)

Jun 2024

- Built full-stack app (TypeScript/React.js + Node.js/FastAPI); integrated GPT-4o for adaptive test generation via clean RESTful API with full session state management, answer validation, per-session score tracking, configurable difficulty parameters, and multi-topic question targeting

LEADERSHIP

Princeton Racket Club

May 2024 - Aug 2024

Tennis Coach and Tournament Director

- Directed tournament logistics, client scheduling, and conflict resolution; coached athletes of all skill levels with customized technique programs and match preparation; managed all end-to-end client communications

Tennis Racket Stringing Services

Jan 2019 - Present

- Founded and scaled to **50+ loyal clients**; manage all operations, pricing, inventory, and client consultations